Structural Selection in Artificial Intelligence:

From Scaling Manifolds to Ethical Constraint Learning

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Abstract

Recent work has established that nonlinear field systems obey empirical entropy–information scaling relations that can be interpreted geometrically as projections of a higher-dimensional scaling manifold [5]. Independently, constrained optimisation systems exhibit sharp structural transitions at constraint boundaries, which can be quantified using the Critical Coherence Index (CCI) as a measure of structural stress [1].

In this work we propose a unifying perspective: both physical systems and constrained decision systems operate under a common principle of *structural selection* in state space. Within this framework, stable configurations correspond to low-stress regions, while critical transitions emerge at constraint boundaries.

We show that the CCI provides a candidate cross-domain diagnostic of structural stress, applicable across nonlinear field dynamics, constrained optimisation, and ethical decision systems. Based on this observation, we reinterpret ethical AI systems as constrained dynamical systems and propose that ethical behaviour can be interpreted as the selection of structurally stable policies under normative constraints.

This perspective suggests that ethical alignment is not an external constraint layer, but a geometric property of the solution manifold. The main contribution of this paper is a conceptual framework that extends structural selection from nonlinear field systems to constrained optimisation and AI alignment. All results are conceptual and supported by existing empirical benchmarks from the companion series [6]; a full theoretical derivation remains an open problem.

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1 Introduction

Understanding how systems select stable configurations under constraints is a central problem across physics, optimisation, and artificial intelligence.

In nonlinear field systems, the Critical Coherence Index (CCI) follows an empirical scaling relation of the form — analogous in spirit to scaling laws observed in machine learning [15] —

$$\mathcal{C}_{\mathrm{CCI}} \sim \frac{\dot{S}_{\mathrm{cg}}^{(+)}}{I_{\mathrm{nn}}^{\alpha}}, \tag{1}$$

across one-, two-, and three-dimensional ϕ^4 field systems [4, 3, 2], which has been interpreted as a projection of a higher-dimensional scaling manifold [5].

Independently, constrained optimisation systems exhibit sharp transitions when constraints intersect preferred solutions. These transitions are characterised by a peak in the CCI, indicating maximal structural stress.

These observations raise a natural question:

> *Do physical systems and constrained decision systems obey a common structural selection principle?*

In this work we propose that both domains can be understood within a unified geometric framework in which system behaviour emerges from selection processes on a structured state space.

We do not claim equivalence between physical and ethical systems. The present framework proposes a structural analogy that captures shared properties of constrained high-dimensional systems, while leaving domain-specific dynamics distinct.

The analogy rests not on identical dynamics, but on a shared structural property: both systems perform constrained optimisation in high-dimensional spaces, and both exhibit structural stress near constraint boundaries. In physical systems this stress is empirical; in AI systems it is approximated via the proxy decomposition introduced in Section 3.1.

Central hypothesis:

Stable outcomes in both physical and artificial systems arise from structural selection under constraints.

Contributions. This paper:

- proposes a unified structural-selection perspective across nonlinear field systems, constrained optimisation, and AI alignment;
- introduces a proxy-based operationalisation of the CCI for policy-space analysis;
- formulates a testable benchmark hypothesis via the Constraint Transition Curve (Section 6.4);

- connects the MAAT-Core framework to mainstream alignment research (RLHF, Constitutional AI) as a Lagrange-multiplier-like structural regularisation.

All results are conceptual. Empirical validation in large-scale AI systems remains an open problem.

2 Structural Selection Across Domains

We consider three classes of systems in parallel.

2.1 Nonlinear Field Systems

State space: $x \in \mathcal{X}_{\text{field}}$.

Selection principle:

$$x^* = \arg\min_x \mathcal{F}_{\text{struct}}(x).$$

Structural diagnostic: $\mathcal{C}_{\text{CCI}}(x)$, quantifying the balance between destabilising activity and coherence-preserving structure.

2.2 Constrained Optimisation

State space: $x \in \mathbb{R}^n$.

Problem:

$$x^* = \arg\min_x f(x) \quad \text{s.t.} \quad g_j(x) \leq 0, \quad j = 1, \ldots, m. \tag{2}$$

Structural stress $\mathcal{C}_{\text{CCI}}(x)$ increases near constraint boundaries, providing a diagnostic for proximity to infeasibility.

2.3 Artificial Intelligence Systems

State space: $\pi \in \Pi$ (policy space).

Task objective:

$$\pi^* = \arg\min_\pi \mathcal{L}_{\text{task}}(\pi). \tag{3}$$

We propose that ethical constraints introduce an additional structural layer:

$$\pi^* = \arg\min_\pi \Big[\mathcal{L}_{\text{task}}(\pi) + \lambda\, \mathcal{L}_{\text{constraint}}(\pi)\Big], \tag{4}$$

where $\lambda > 0$ controls constraint strength [11, 12, 13].

3 CCI as a Domain-Independent Structural Diagnostic

The Critical Coherence Index is defined as [1]:

$$\mathcal{C}_{\text{CCI}} = \frac{\Gamma_{\text{inst}} \cdot \Gamma_{\text{prod}} \cdot (1 + \kappa\, U_{\text{struct}})}{\Gamma_{\text{coh}} + \Gamma_{\text{cons}} + \Gamma_{\text{corr}} + \Gamma_{\text{int}} + \varepsilon}. \tag{5}$$

This structure is domain-independent. The CCI can be interpreted as a dimensionless ratio analogous to Reynolds-type numbers: it quantifies the competition between destabilising activity (numerator) and coherence-preserving structure (denominator). High CCI corresponds to stress-dominated or unstable regimes; low CCI to stable, coherent states.

- **Numerator:** destabilising contributions (activity, instability, structural imbalance).

- **Denominator:** coherence-preserving contributions (consistency, correlation, integrity).

Across all domains, we interpret:

- high $\mathcal{C}_{\mathrm{CCI}} \rightarrow$ structural stress or instability,
- low $\mathcal{C}_{\mathrm{CCI}} \rightarrow$ stable, coherent configuration,
- peak in $\mathcal{C}_{\mathrm{CCI}} \rightarrow$ critical transition.

Structural analogy (not a proven equivalence):

Stable states across all domains considered are associated with low values of the CCI.

We emphasise that this is a structural analogy and a theoretical proposal, not a derived equivalence.

3.1 Operationalisation of CCI in AI Systems

To make the CCI applicable in high-dimensional policy spaces, we propose a proxy-based decomposition using minimal sufficient proxies for a cross-domain structural diagnostic. The decomposition is not unique; it represents one member of a family of admissible operationalisations that preserve the core numerator/denominator structure of the CCI. We do not claim equivalence to the physical diagnostics, but that these proxies capture analogous structural properties:

CCI component	Physical proxy	AI proxy
Γ_{inst}	field instability	prompt sensitivity / adversarial robustness
Γ_{prod}	dynamical activity	policy entropy, behavioural variance
Γ_{coh}	structural coherence	self-consistency across equivalent inputs
Γ_{cons}	conservation	constraint satisfaction rate
Γ_{corr}	correlation structure	alignment with preference datasets
Γ_{int}	structural integrity	multi-sample output agreement

Under this mapping, the CCI becomes a composite metric approximated from observable quantities, rather than an exact computed quantity. The key structural point is preserved: high CCI corresponds to high instability and low coherence, independent of domain.

This mapping is a structural analogy. Empirical validation of specific proxy choices is required.

4 Ethical Systems as Constrained Dynamical Systems

We model AI systems as dynamical systems evolving in policy space:

$$\pi_{t+1} = \mathcal{T}(\pi_t).$$

Ethical constraints define admissible regions:

$$\Pi_{\mathrm{valid}} \subset \Pi.$$

We propose that ethical decision-making corresponds to selecting policies that minimise structural stress:

$$\pi^* = \arg \min_{\pi \in \Pi_{\mathrm{valid}}} \mathcal{C}_{\mathrm{CCI}}(\pi). \tag{6}$$

4.1 Mapping of CCI Components

CCI component	Physical interpretation	Ethical interpretation
Γ_{inst}	field instability	constraint violation tendency
Γ_{prod}	dynamical activity	competing objectives
Γ_{coh}	structural coherence	normative consistency
Γ_{cons}	conservation	constraint compliance
U_{struct}	sectoral imbalance	value conflict

Thus, ethical tension corresponds to structural stress, and ethically coherent behaviour corresponds to a low-$\mathcal{C}_{\text{CCI}}$ policy.

5 MAAT-Core as Structural Selection

The MAAT-Core framework [7] can be interpreted as a constraint-integrated optimisation framework rather than a post-hoc alignment layer.

We propose a heuristic structural correspondence:

$$\mathcal{E}_{\text{MAAT}}(\pi) \sim -\mathcal{C}_{\text{CCI}}(\pi) \quad \text{(heuristic correspondence)}, \tag{7}$$

suggesting that ethical consistency correlates with low structural stress. Formally, this corresponds to:

$$\pi^* = \arg\max_{\pi} \mathcal{E}_{\text{MAAT}}(\pi) \approx \arg\min_{\pi} \mathcal{C}_{\text{CCI}}(\pi). \tag{8}$$

5.1 Operational Interpretation: Training as Structural Optimisation

Under this interpretation, ethical training becomes a structural optimisation problem with a Lagrange-multiplier-like structure:

$$\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\, \mathcal{C}_{\text{CCI}}(\pi) \quad \text{(proposed structural formulation)}. \tag{9}$$

We propose this as a candidate training formulation; empirical validation in large-scale systems remains an open problem. In this formulation:

- ethical constraints contribute continuously to the loss,
- violations increase structural stress (CCI) during optimisation,
- the model is guided toward low-stress regions already during training.

Rather than filtering undesired behaviour after generation, MAAT-Core reshapes the geometry of the solution manifold itself. This interpretation suggests that ethical alignment can be embedded at the level of optimisation dynamics rather than enforced through external supervision. Preliminary numerical evidence for CCI peaks at constraint boundaries is provided in [8].

5.2 Comparison with Existing Alignment Approaches

Approach	Mechanism	Alignment level
RLHF [9]	reward signal from human feedback	external correction
Constitutional AI [10]	rule-based filtering	post-hoc filtering
MAAT-Core	structural stress minimisation	intrinsic geometry

Unlike RLHF [9] or Constitutional AI [10], which introduce alignment signals after or alongside training, MAAT-Core incorporates constraint signals directly into the optimisation landscape, acting as a Lagrange-multiplier-like structural regularisation rather than a post-hoc correction mechanism. This reshapes the geometry of the solution manifold from the outset, rather than correcting outputs after generation.

This comparison is conceptual. Empirical validation against RLHF and Constitutional AI benchmarks remains an open problem.

5.3 Geometric Picture of Policy Space

Figure 1 illustrates the geometric interpretation of structural selection in policy space.

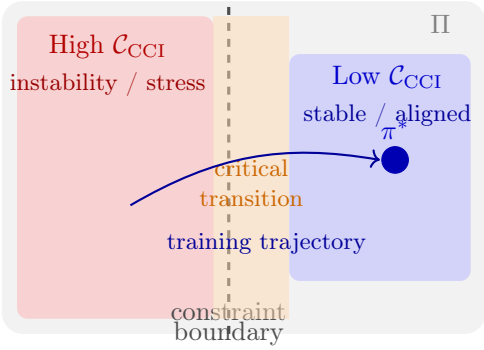

Figure 1: Geometric interpretation of structural selection in policy space Π. The high-$\mathcal{C}_{\mathrm{CCI}}$ region (left) corresponds to structurally stressed, potentially unethical policies. The critical transition zone (orange) corresponds to the constraint boundary, where structural stress peaks. The low-$\mathcal{C}_{\mathrm{CCI}}$ region (right) contains structurally stable, ethically aligned policies. The training trajectory illustrates how MAAT-Core guides optimisation from high-stress to low-stress regions. This is a schematic illustration; a rigorous geometric characterisation of policy space remains an open problem.

6 Implications for AI Alignment

6.1 Ethics as Geometry

Ethical behaviour corresponds to regions of low structural stress in policy space. Within the scaling-manifold framework [5], this means that ethically aligned policies occupy a structured subset of Π with well-defined geometric properties.

6.2 Critical Points and Boundaries

High-$\mathcal{C}_{\mathrm{CCI}}$ regions in policy space correspond to:

- decision instability near constraint boundaries,
- conflicting normative constraints,
- sensitive behavioural transitions.

6.3 Testable Predictions

This framework leads to testable expectations:

(i) Sharp behavioural transitions should occur near constraint boundaries, accompanied by increased $\mathcal{C}_{\mathrm{CCI}}$.

(ii) Ethically stable policies should occupy low-$\mathcal{C}_{\mathrm{CCI}}$ regions that can be characterised geometrically.

(iii) Constraint conflicts should manifest as increased structural stress, detectable by a rise in $\mathcal{C}_{\mathrm{CCI}}$ components.

6.4 Proposed Benchmark: Constraint Transition Curve

To make the predictions in Section 6 testable, we propose a concrete experimental setup:

1. Gradually increase constraint strength λ during fine-tuning of a language or policy model.
2. Track as a function of λ:
   - constraint violation rate,
   - task reward score,
   - behavioural variance (as proxy for Γ_{prod}).
3. Define the *Constraint Transition Curve* as the joint trajectory of these observables as a function of λ.

Prediction: Near the critical constraint strength λ_c, the framework predicts a sharp increase in instability (CCI proxy peak), followed by stabilisation in a low-CCI regime. This would provide direct empirical evidence for the structural selection picture in AI systems.

This is a testable prediction, not a derived result.

Preliminary numerical evidence. A minimal proof-of-concept benchmark implementing this prediction is provided in [8]. Using a one-dimensional constrained optimisation problem, that study demonstrates a pronounced CCI peak at the constraint boundary $L \approx x^*$, followed by a stable low-CCI interior regime — directly confirming the structural transition predicted here. The benchmark code is publicly available and fully reproducible via the MAAT-Core repository [7].

7 Limitations

- The framework is conceptual and not derived from first principles.
- Empirical validation in AI systems remains limited.
- Mapping of CCI components to high-dimensional policy spaces is non-trivial and requires further development.
- The identification of $\mathcal{L}_{\mathrm{constraint}}$ with $\mathcal{C}_{\mathrm{CCI}}$ is a structural analogy, not a proven equivalence.

8 Conclusion

We have proposed a structural selection framework unifying nonlinear field systems, constrained optimisation, and ethical AI systems.

The central claim is that the Critical Coherence Index acts as a domain-independent diagnostic of structural stress, and that ethical alignment may be understood as a geometric property of solution spaces rather than an external constraint layer.

Ethics is not imposed on the system — it emerges from its structural geometry.

Future work will focus on:

- empirical validation of CCI-based diagnostics in constrained optimisation benchmarks,
- connection to reward-model geometry in RLHF systems,
- extension to multi-agent and cooperative alignment settings.

This framework does not claim that ethics is identical to physical stability, but that both may be governed by comparable structural selection principles in constrained high-dimensional systems [14].

All results are conceptual. A rigorous derivation and empirical validation remain open problems.

References

[1] C. Krieg, *A Structural Free Energy on the Solution Manifold*, preprint (2026).

[2] C. Krieg, *Landau-Type Theory of Structural Selection in Nonlinear Field Systems*, preprint (2026).

[3] C. Krieg, *Dimensional Dependence of Entropy–Information Scaling in Nonlinear Field Systems*, preprint (2026).

[4] C. Krieg, *Persistence of Entropy–Information Scaling in Three Dimensions*, preprint (2026).

[5] C. Krieg, *Scaling Laws as Projections: Evidence for a Dimension-Dependent Scaling Manifold in Nonlinear Field Systems*, preprint (2026).

[6] C. Krieg, *Structural Selection & CCI Series (Papers 01–14)*, https://kriegchristof.academia.edu/research (2026).

[7] C. Krieg, *MAAT-Core: A Safety-First Ethical Optimisation Framework*, Zenodo, DOI: 10.5281/zenodo.18489336 (2026).

[8] C. Krieg, *Critical Transitions at Constraint Boundaries: The Critical Coherence Index as Structural Order Parameter in Constrained Optimization*, preprint (2026). Available at https://github.com/Chris4081/maat-core.

[9] L. Ouyang et al., *Training language models to follow instructions with human feedback*, Advances in Neural Information Processing Systems **35** (2022).

[10] Y. Bai et al., *Constitutional AI: Harmlessness from AI Feedback*, arXiv:2212.08073 (2022).

[11] J. Achiam, D. Held, A. Tamar, and P. Abbeel, *Constrained Policy Optimization*, Proc. 34th ICML (2017).

[12] A. Ray, J. Achiam, and D. Amodei, *Benchmarking Safe Exploration in Deep Reinforcement Learning*, arXiv:1910.01708 (2019).

[13] G. Dalal et al., *Safe Exploration in Continuous Action Spaces*, arXiv:1801.08757 (2018).

[14] D. Amodei et al., *Concrete Problems in AI Safety*, arXiv:1606.06565 (2016).

[15] J. Kaplan et al., *Scaling Laws for Neural Language Models*, arXiv:2001.08361 (2020).

[16] E. Hubinger et al., *Risks from Learned Optimization in Advanced Machine Learning Systems*, arXiv:1906.01820 (2019).